

✓ Introduction of DOS (Disk Operating System)

What is DOS?

DOS stands for **Disk Operating System**.

It is a type of **operating system** that was used before Windows became popular. DOS is based on a **Command-Line Interface (CUI)**, which means users have to **type commands** using the keyboard to perform any task — such as opening files, creating folders, or copying data.

Unlike modern systems like Windows or macOS that show **icons and buttons** (GUI), DOS only shows **text on the screen**, and you control the computer by typing specific commands.

♦ History and Development

- DOS was developed by **Microsoft** in the year **1981**.
- It was made for **IBM personal computers (PCs)**.
- The first version was called **MS-DOS (Microsoft Disk Operating System)**.
- Before graphical operating systems were introduced, DOS was the **main operating system** used in homes, schools, and offices.

♦ Why is it called "Disk Operating System"?

Because:

- It runs from a **disk** (like a floppy disk or hard disk).
- It helps in **managing files** and **folders** stored on the disk.
- It provides a way to access and use the data stored in the **computer's drives**.

♦ Basic Characteristics of DOS

Feature	Description
Text-based Interface	You interact with DOS by typing commands. No graphics.
Single-user OS	Only one user can use the system at a time.
Single-tasking OS	It can do only one task at a time.
Small size	Uses very little memory and resources.
Command-line based	You control everything by writing commands manually.

♦ Main Uses of DOS (Earlier Days)

- Managing files and folders
- Formatting disks
- Running programs (EXE, COM, bat files)
- Checking and fixing errors in storage
- Creating scripts using batch files

♦ Why Learn DOS Today?

Even though DOS is not commonly used today, learning it helps in:

- Understanding how computers work at the basic level
- Learning about file and memory management
- Getting introduced to **command-line tools**, which are still used in Linux, Mac, and even Windows (CMD/PowerShell)

♦ Example of a DOS Command:

bash

DIR

This command shows all files and folders in the current directory.

Loading DOS (Disk Operating System)



What is “Loading DOS”?

Loading DOS means **starting the DOS operating system** when you switch on the computer. It is the process by which the computer prepares itself to work with DOS and gets ready to take your commands.

Just like you need to wake up and get ready in the morning, a computer also needs to “wake up” and load the operating system — and that process is called **booting** or **loading DOS**.

Steps Involved in Loading DOS:

1 Power On the Computer

- When you press the power button, the computer turns on and starts checking its basic parts.
- This is called **POST (Power-On Self Test)**.
- It checks RAM, keyboard, and other basic hardware.

2 BIOS is Activated

- **BIOS (Basic Input/Output System)** is a small program stored on the motherboard.
- It checks which drive (Hard Disk, Floppy, or USB) has the DOS system files.
- BIOS finds the drive where DOS is stored and starts to load it.

3 Loading System Files

DOS loads three important system files into memory:

File Name	Description
IO.SYS	Controls input/output devices like keyboard, screen
MSDOS.SYS	Main operating system core
COMMAND.COM	Loads internal DOS commands and shows the prompt

♦ These files must be present in the **root directory** of the bootable drive (like C:\ or A:\) for DOS to work properly.

4 Displaying System Prompt

- After loading these files, DOS shows a **system prompt**, like:

C:\>

This means DOS is ready, and now you can **type commands** to do your work.

📌 What If DOS Files Are Missing?

- If system files like IO.SYS or COMMAND.COM are missing or damaged, the computer will show an error:



Topic 3: System Prompt in DOS

◆ What is a System Prompt?

A **System Prompt** is the **blinking cursor line** in DOS that shows the system is **ready to accept commands**.

It appears on the screen after DOS has successfully loaded.

✓ It is the starting point for all your actions in DOS.

◆ How Does It Look?

• C:\>

Symbol Meaning

C: Current drive (C drive)

\ Current directory (Root folder)

> System is ready for your command

This prompt tells you:

"You are now in the C drive, at the root folder. I'm ready — type your command."

List of Common DOS Commands You Can Use at the System Prompt

Below are categorized commands with their **purpose** and **examples**.

File and Directory Commands

Command	Description	Example
DIR	Displays list of files and folders	DIR
CD	Changes the directory	CD STUDENTS
CD..	Goes back one level	CD..
MD / MKDIR	Creates a new folder (directory)	MD PROJECT
RD / RMDIR	Deletes an empty folder	RD PROJECT
COPY	Copies file from one place to another	COPY A.TXT B.TXT
DEL	Deletes a file	DEL A.TXT
REN	Renames a file	REN A.TXT B.TXT
TYPE	Displays content of a text file	TYPE NOTE.TXT

Disk and Drive Commands

Command	Description	Example
A: / B:	Switch to floppy drive A:	
D:	Switch to D drive	D:
FORMAT	Formats a drive	FORMAT A:
CHKDSK	Checks disk for errors	CHKDSK

System and Screen Commands

Command	Description	Example
CLS	Clears the screen	CLS
VER	Displays DOS version	VER
DATE	Views or sets the system date	DATE
TIME	Views or sets the system time	TIME
EXIT	Exits from DOS or command window	EXIT

Batch Files and Automation

Command / File	Description	Example
.BAT files	Contains a list of commands; runs automatically	RUN.BAT

Command / File	Description	Example
AUTOEXEC.BAT	Special file that runs commands at startup	Auto executes



Drive and Its Operations in DOS

♦ What is a Drive?

A **drive** is a **storage location** in your computer where data, files, and programs are stored. In DOS, each drive is represented by a **letter followed by a colon**.

📌 Examples of Drives:

- C: – Hard Disk (main storage)
- D: – CD/DVD drive or second hard drive
- E:, F: – USB drives or additional partitions
- A: and B: – Floppy disk drives (older systems)

♦ Types of Drives

Drive Letter	Description
C:	Primary hard disk drive
D:	Secondary hard drive or optical drive
A: / B:	Floppy disk drives (rare now)
E: onwards	USB, external drives, network drives

♦ How to Change Drives in DOS

To switch from one drive to another, just **type the drive letter and colon** and press **Enter**.

✅ Example:

```
C:\> D:
D:\> E:
```

♦ Drive Operations in DOS

DOS allows many basic operations on drives using simple commands.

📁 1. Viewing Drive Contents

Command: DIR

Use: Lists all files and folders on the current drive.

Example:

```
C:\> DIR
```

2. Formatting a Drive

Command: FORMAT

Use: Erases all data on the drive and prepares it for use.

Example:

```
A:\> FORMAT A:
```

 **Caution:** Formatting deletes everything on the drive.

3. Checking Disk for Errors

Command: CHKDSK (Check Disk)

Use: Checks the drive for file system errors and displays disk information.

Example:

```
C:\> CHKDSK
```

4. Displaying System Date and Time

These commands also work across all drives.

- **Command:** DATE → Shows/sets system date
- **Command:** TIME → Shows/sets system time

Example:

```
C:\> DATE
```

```
C:\> TIME
```

5. Exiting Drive or Returning to Root

- **Command:** CD\ → Returns to root directory

- **Command:** EXIT → Closes the DOS window or exits command prompt

In Simple Words:

A **drive** is like a separate **cupboard** inside your computer.

- You can **open** it (by changing the drive),
- **see what's inside** (with DIR),
- **clean it completely** (FORMAT),
- and **check for problems** (CHKDSK).

Summary Table of Drive Commands

Command	Description	Example
C:	Change to C drive	C:
D:	Change to D drive	D:
DIR	List files/folders	DIR
FORMAT	Format drive	FORMAT A:
CHKDSK	Check drive errors	CHKDSK
EXIT	Exit command prompt	EXIT

Introduction to File and Directory in DOS

♦ What is a File?

A **file** is a **collection of related data or information**, stored in digital form, which can be created and used by users or applications.

 You can think of a file like a **digital paper** where something is written or saved — like a document, photo, video, or program.

Example of Files:

- notes.txt – A text file
- photo.jpg – An image file
- project.docx – A document file
- game.exe – A program file

♦ What is a Directory?

A **directory** is like a **folder** that contains files or even other directories (called subdirectories). Directories help in **organizing files** properly so they're easy to find and manage.

 Example:

Imagine a **folder named "College"** on your computer. Inside it, you keep files like:

- assignment.docx
- result.pdf
- A subfolder named Semester1 with more files in it.

♦ Types of Files

File Type	Extension	Example
Text File	.txt	info.txt
Document File	.docx	resume.docx
Image File	.jpg	photo.jpg
Program File	.exe	game.exe

♦ DOS File Naming Rules

1. **File Name:** Up to 8 characters
2. **Extension:** 3 characters (optional), separated by a dot
3. **No spaces allowed** in file names
4. **No special symbols** like *, ?, /, \, :, etc.

 **Example of Valid File Name:**

project.txt

notes.doc

 **Invalid File Name:**

my project!.txt

♦ Types of Directories in DOS

Type	Description
Root Directory	The main (top-level) directory of a drive (like C:\)
Subdirectory	A directory inside another directory
Current Directory	The directory you are working in
Parent Directory	The directory one level above the current directory

♦ DOS Commands to Manage Files & Directories

Creating a Directory

Command: MD or MKDIR

Use: Creates a new folder/directory.

Example:

```
C:\> MD STUDENT
```

Deleting a Directory

Command: RD or RMDIR

Use: Removes an empty folder.

Example:

```
C:\> RD STUDENT
```

Creating a File

Command: COPY CON filename

Use: Creates a new file and allows you to type its content.

Example:

```
C:\> COPY CON HELLO.TXT
```

```
Hello students!
```

```
^Z  ← (Press Ctrl + Z to save)
```

Displaying Directory Contents

Command: DIR

Use: Lists all files and directories.

Example:

```
C:\> DIR
```

Renaming Files

Command: REN oldname newname

Use: Renames a file.

Example:

```
C:\> REN HELLO.TXT GREETING.TXT
```

Deleting Files

Command: DEL filename

Use: Deletes a file.

Example:

```
C:\> DEL GREETING.TXT
```

Difference Between Internal and External Commands in DOS

Feature	Internal Commands	External Commands
Storage Location	Stored inside COMMAND.COM	Stored as separate files (.EXE or .COM)

Feature	Internal Commands	External Commands
Loading	Loads automatically when DOS starts	Loaded from disk only when the command is used
Availability	Always available in memory	Available only if the file exists in the disk/PATH
Execution Speed	Faster because already in memory	Slightly slower because it needs to load first
Memory Usage	Stays in memory permanently	Uses memory temporarily while running
Error if File Missing	No error, always works	Shows “Bad command or file name” if missing
Purpose	Used for basic and common tasks	Used for advanced or special tasks
Examples	DIR, CLS, COPY, DEL, CD, TIME, DATE	FORMAT, CHKDSK, DISKCOPY, TREE, XCOPY

Batch File in DOS

A **Batch File** is a **text file** that contains a series of **DOS commands**.

- These commands are **executed automatically** in the order they are written when the batch file is run.
- Batch files help to **perform repetitive tasks quickly** without typing commands again and again.

Key Points about Batch Files

1. **File Extension:**
 - A batch file always has the **.BAT** extension.
 - Example: MYTASK.BAT
2. **Execution:**
 - When you **double-click** the batch file or **type its name in DOS**, all commands inside it **run one by one automatically**.
3. **Purpose:**
 - Batch files are used to **save time** by running **multiple commands together**.
 - Useful for **automation** of daily tasks like creating backups, cleaning files, or starting programs.

Example of a Batch File

Let's create a simple batch file step by step:

Step 1: Open **Notepad** and type the commands:

```
@echo off  
echo Hello, Welcome to DOS  
dir  
pause  
cls  
echo Task Completed!
```

Step 2: Save the file as:

welcome.bat

Step 3: Run the batch file by:

- Double-clicking it in Windows, or
- Typing in DOS:

C:\>welcome

What Happens:

1. Shows the message "Hello, Welcome to DOS"
2. Displays the list of files in the current folder (DIR)
3. Waits for a key press (PAUSE)
4. Clears the screen (CLS)
5. Shows the message "Task Completed!"

Common Commands Used in Batch Files

1. **ECHO** – Displays messages.
2. **PAUSE** – Waits for the user to press any key.
3. **CLS** – Clears the screen.

4. **REM** – Adds comments in the batch file (ignored by DOS).
5. **GOTO** – Jumps to a specific label in the file.
6. **IF** – Executes commands based on a condition.

Advantages of Batch Files

1. **Saves Time** – Automates repetitive tasks.
2. **Easy to Use** – Can be created using Notepad.
3. **No Programming Needed** – Only basic DOS commands are required.
4. **Automation** – Useful for backups, file organization, or system checks.

Difference between CUI and GUI

Point	CUI (Character User Interface)	GUI (Graphical User Interface)
1. Meaning	Interaction through text commands	Interaction through graphics (icons, menus, windows)
2. Ease of Use	Difficult, needs command knowledge	Easy and user-friendly
3. Input Device	Only keyboard	Keyboard, mouse, touch, etc.
4. Speed	Faster for experts	Slower due to graphics
5. Resources	Needs less memory and processing power	Needs more memory and processing power
6. Multitasking	Limited	Supports multitasking with multiple windows
7. Examples	MS-DOS, UNIX command line, CMD	Windows, macOS, Linux (Ubuntu GUI), Android, iOS

👉 In short:

- **CUI = Text & Commands**
- **GUI = Graphics & Icons**